

AHMAD SYAFIQ KAMIL

Senior Software Engineer | DevOps & Cloud Specialist

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Senior Software Engineer with 7+ years of experience architecting scalable distributed systems, cloud-native infrastructure, and full-stack web applications. Expert in bridging legacy government systems with modern tech stacks (Docker, Kubernetes, Next.js). Proven track record of deploying high-availability services for international diplomatic missions and managing national-level research data pipelines. Currently expanding expertise into Web3 and Blockchain development (Solidity, Rust).

TECHNICAL SKILLS

Languages: Python (Advanced), JavaScript/TypeScript, Rust, Solidity, PHP, SQL

Backend & Frameworks: Django REST Framework, Node.js, Next.js, Flask

DevOps & Cloud: Docker, Kubernetes (K8s), CI/CD Pipelines, DigitalOcean, Linux Administration, Nginx

Data & AI: MLOps, Computer Vision (OpenCV), Pandas, NumPy, ODK Central

Tools: Git, Postman, Figma, Jira, n8n

PROFESSIONAL EXPERIENCE

Syamil Solusindo Semesta | Remote / Jakarta

Senior Full Stack Engineer & Lead Consultant | 2022 – Present

Led the digital transformation of Indonesian diplomatic missions, focusing on system reliability and modernization.

- **System Architecture:** Architected and deployed a microservices-based "E-Office" platform using Next.js and Django, currently serving 500+ daily active users across KBRI Malaysia and KJRI Dubai.
- **DevOps Implementation:** Migrated legacy on-premise servers to cloud infrastructure using Docker containers, achieving 99.9% uptime and reducing maintenance downtime by 40%.
- **Database Optimization:** Unified 5+ disparate diplomatic databases into a single MySQL cluster, optimizing query performance and enabling real-time data synchronization between missions.

- **Security:** Implemented role-based access control (RBAC) and encrypted data transmission standards for sensitive diplomatic communications.

Badan Riset dan Inovasi Nasional (BRIN) | Jakarta

Research Engineer & Data Specialist | 2022 – Present

Spearheaded data engineering initiatives to support national research infrastructure.

- **Data Pipeline Automation:** Engineered automated data collection pipelines using Python and ODK Central, processing 500+ heterogeneous datasets annually for national research repositories.
- **Tooling Development:** Developed internal Python scripts to scrape, clean, and visualize secondary data sources, reducing manual data entry workload by 20%.
- **Technical Validation:** Collaborated with cross-functional teams to validate technical feasibility of digital infrastructure recommendations before implementation.
- **System Integration:** Built REST APIs to integrate research data systems with national databases, ensuring seamless data flow between institutions.

Upwork (Freelance) | Remote

Full Stack Developer & DevOps Engineer | 2023 – Present

- **Infrastructure as Code:** Deployed self-hosted ODK Central instances on DigitalOcean Droplets using Docker Compose for international NGO clients conducting medical surveys.
- **High Availability:** Configured load balancers and automated backup scripts, ensuring 95%+ uptime and data integrity for critical field research applications.
- **Client Support:** Maintained a 90%+ first-response resolution rate for technical troubleshooting, delivering comprehensive documentation to reduce onboarding time by 25%.

Badan Pengkajian dan Penerapan Teknologi (BPPT) | Jakarta

AI/ML Engineer | 2021 – 2022

- **MLOps Deployment:** Designed and implemented MLOps pipelines on national research servers using Kubernetes and Docker, enabling scalable deployment of Face Recognition and Pattern Recognition models.
- **Computer Vision:** Developed Android applications integrated with backend Computer Vision APIs using Python and OpenCV for real-time image processing in field data collection.
- **Process Optimization:** Improved model deployment efficiency by 30% through implementation of CI/CD automation pipelines and containerization best practices.
- **Knowledge Transfer:** Trained 50+ engineers on scalable AI infrastructure and containerization technologies.

Embassy of Indonesia in Kuala Lumpur (KBRI) | Malaysia

Software Engineer | 2019 – 2021

- **Full Lifecycle Development:** Built and deployed 3 mission-critical applications (Diplomatic Letter Management System, Queue Management System, COVID-19 Aid Distribution System) within 14 months using PHP, Django, and PostgreSQL.
- **High Traffic Management:** System handled 100+ diplomatic letters daily and managed high-concurrency requests during pandemic relief operations serving thousands of Indonesian citizens.
- **Legacy System Modernization:** Digitized manual workflows, reducing processing time by 30% and improving service delivery for embassy staff and citizens.
- **Infrastructure Maintenance:** Maintained 10+ embassy IT systems and networks with 99% uptime while coordinating with external vendors.

Universitas Nahdlatul Ulama | Surabaya

Lecturer (Computer Science) | 2018 – 2019

- Taught 24 credit hours weekly across core CS courses: Algorithms, Data Structures, Web Development, and Mobile Programming.
- Mentored 100+ students per semester with measurable learning outcomes and curriculum aligned to industry standards.

KEY PROJECTS

BetterFuture & E-Voting DApps (2025)

Developed decentralized applications using Solidity smart contracts and Vite.js frontend. Implemented secure voting mechanisms with on-chain verification and tokenized incentive systems for community governance.

E-Notes Blockchain (2025)

Built a decentralized note-taking application with end-to-end encryption using Solidity, Next.js, Flask, and Docker. Integrated IPFS for distributed storage.

Parvata & FOCO iOS Apps (2024)

Native iOS development using Swift and SwiftUI. Built productivity and focus management tools with CoreData integration, background notifications, and intuitive user interfaces following Apple HIG.

N8N-Grok Automation (2025)

Deployed self-hosted n8n workflow automation platform using Docker for AI-powered content generation and API integrations.

Consular Service Integration Platform

Created a unified API gateway connecting scattered consular databases across 5+ Indonesian missions in Malaysia, enabling single-sign-on (SSO), centralized data management, and real-time synchronization.

MLOps Infrastructure for National Research

Architected shared machine learning infrastructure using Kubernetes and Kubeflow, adopted by multiple research centers for deploying computer vision and pattern recognition models at scale.

EDUCATION

Master of Computer (M.Kom) in Information Systems

Institut Teknologi Sepuluh Nopember (ITS) Surabaya | GPA 3.38 | 2015–2018

Bachelor of Computer (S.Kom) in Information Systems

Universitas Jember | GPA 3.34 | 2010–2015

CERTIFICATIONS & TRAINING

- SUI Blockchain Bootcamp (2025)
- Blockchain Developer (Solidity) – BlockDevId (2025)
- Langchain Academy: Deep Agent with Langgraph (2025)
- Apple Academy Developer (2024)
- Digitalent Big Data (2022)
- Lead Auditor ISO 27001:2013 – Information Security Management (2019)

LANGUAGES

English: B2 Upper Intermediate (Professional Working Proficiency)

Bahasa Indonesia: Native